

Imputing missing electroencephalography data using graph signal processing

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ABSTRACT

Graph Signal Processing (GSP) is a framework for analyzing signals defined over a graph. Considering the electrodes used to record the electroencephalogram (EEG) as a sensor network makes it possible to use GSP to analyze EEG signals. Using the graph over which the signal is defined allows one to take advantage of a signal structure that is ignored by classic signal processing approaches. However, there are many details about how to use GSP to analyze the EEG that are not studied in the literature. Here we show an example of how to impute missing EEG data using GSP. We show that GSP allows reconstructing EEG missing data with a lower error than a classic approach based on radial basis functions, confirming that the underlying graph over a graph over which the signal is defined contains relevant information that can be exploited to improve a given signal processing task. By studying two approaches for building the graph (k -nearest neighbors and a thresholded Gaussian kernel) and the effect of its parameter, we highlight the importance of building the graph appropriately. These results show the potential of incorporating GPS techniques into the EEG processing pipeline.

Keywords: Electroencephalography, graph signal processing, missing data

1. INTRODUCTION

The electroencephalogram (EEG) is the recording of the electric potentials over the scalp. These electric potentials are the result, mainly, of the brain cortex activity,¹ and thus, the EEG can be used to infer the state of this part of the body.² The EEG is a non-invasive low-risk technique used in clinical and research settings. Clinical practitioners use the EEG to monitor the depth of anesthesia, diagnose epilepsy and sleep disorders, to continuous monitoring of intensive care unit patients, among others.^{3–5} Researchers use the EEG in studies related to neuroscience, experimental psychology, and neurolinguistics, among others.⁶ The EEG has also been used in other settings, including real-time detection of driver fatigue⁷ and consumer applications.⁸

On the other hand, *Graph Signal Processing* (GSP)⁹ is framework that extends signal processing operations—filtering, interpolation, wavelet analysis, etc.—to signals defined over a graph, thus, taking advantage of existing data’s structure that otherwise can not be exploited. To acquire the EEG one places a set of electrodes over the subject’s scalp on a predefined set of locations. In this work we show how GSP can be used to process the EEG signal by considering these electrodes as a sensor network and that this sensor network induces a graph with an associated signal. In particular, we show how GSP can be used to improve the performance of classic EEG data imputation techniques. Data imputation is a common situation in practice, that happens when the skin-electrode connection is inappropriate. Under these circumstance, the signal of one or more electrodes is missing, and one needs to impute the missing data using the measurements from the remaining electrodes. Our results show that GPS-based signal imputation is better than the classic approach.

Section 2 gives a brief introduction to GSP. In section 3 we review how GSP have been used in neuroimaging in general and in EEG in particular. Then, section 4 describes the missing EEG data imputation experiment and its result. Section 5 finishes the manuscript with the discussion and conclusion.

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2. GRAPH SIGNAL PROCESSING

The foundations of GSP can be traced back to Püschel and Moura's work on algebraic signal processing¹⁰ in 2003 and the work of several researchers on graph Laplacian spectral clustering and image processing. For a detailed historical perspective, see the Introduction, section B, of Ortega et al.⁹ In this work, we use the setting defined by Shuman,¹¹ which has the advantage that generalizes the standard finite length discrete time signal processing case.* This setting considers a weighted[†] graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, W\}$, where $\mathcal{V}$ is the set of vertices (with cardinality $|\mathcal{V}| = N$), $\mathcal{E}$ is a set of edges, and W is a weighted adjacency matrix. The cornerstone of GSP is the graph Laplacian matrix[‡]

$$\mathcal{L} = D - W,$$

where D is the diagonal degree matrix with entry d_i equals to the sum of the weights incident to vertex i .¹² Since the graph Laplacian matrix is symmetric with real entries, it has non-negative eigenvalues. We denote this set of eigenvalues as

$$\sigma(\mathcal{L}) = \{\lambda_0, \dots, \lambda_{N-1}\}.$$

The set $\sigma(\mathcal{L})$ is called the spectrum of the graph Laplacian. One of the properties of the graph Laplacian is that its smallest eigenvalue is 0.¹² We also assume that the eigenvalues are ordered. Thus, we can write

$$0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{N-1}.$$

Also, because the graph Laplacian matrix is symmetric with real entries, its eigenvectors are orthonormal.¹² Although the eigenvectors are not unique, we fix them for a given problem. We denote the set of eigenvectors by $\{u_\ell\}_{\ell=0,1,\dots,N-1}$.

The eigenvectors of the graph Laplacian provide the notion of frequency, with eigenvectors associated with larger eigenvalues corresponding to higher frequencies.¹³ For instance, for any graph, the first eigenvector u_0 with eigenvalue $\lambda_0 = 0$ is always a constant. Thus, this eigenvector matches the concept of a continuous or DC component. Another example useful to illustrate the relationship between the eigenvectors of the graph Laplacian and the concept of frequency is the case of the path (or line) graph. In this graph, vertex i is connected only to vertex $i-1$ and $i+1$ (except vertex 0 and $N-1$, which are connected only to vertex 1 and $N-2$, respectively). The eigenvectors of this graph are sinusoidal, with frequencies increasing with the corresponding eigenvalue.^{14–16} A signal defined over a path graph matches the standard finite-length signal considered in classical signal processing. Thus, the sinusoidal nature of the eigenvectors of the path graph Laplacian allows one to consider GSP as a generalization of the classic signal processing framework.

These properties of the graph Laplacian allow the definition of a *graph Fourier transform* for a signal $f : \mathcal{V} \rightarrow \mathbb{R}$ defined over the vertices of the graph. The graph Fourier transform $\hat{f}(\lambda_\ell)$ and its corresponding inverse are given by

$$\hat{f}(\lambda_\ell) = \sum_{i=0}^{N-1} f(i) u_\ell^*(i), \quad f(i) = \sum_{\ell=0}^{N-1} \hat{f}(\lambda_\ell) u_\ell(i).$$

The justification for this definition is based on the following fact: The classical Fourier transform expands the signal in terms of complex exponentials, which are eigenfunctions of the one-dimensional Laplace operator. Similarly, the graph Fourier transform expands the signal in terms of eigenvalues of the graph Laplacian matrix, a similar operator (hence the name of this matrix).¹¹

Two other critical signal processing ingredients that are extended to the GSP theory are the concept of *shift* or *delay*, and the concept of a filter. For a detailed description about these ideas see the work of Shuman et al., Sandryhaila et al., and Ortega et al., and references therein.^{9,11,17}

Before doing any GSP-based signal processing, one must build an appropriate graph over which the signal is defined. As we will see in section 4, GSP's performance depends on how we define the edge set of the graph. A

*Under this framework, one gets a finite length discrete time signal with a signal defined over an unweighted line graph.

[†]The case of an unweighted graph can be handle as a particular case with all the weights set to one.

[‡]It is also possible to use the normalized graph Laplacian $\tilde{\mathcal{L}} = D^{-1/2} \mathcal{L} D^{-1/2}$. See Shuman et al.¹¹ for a discussion about the implications of using this matrix instead of the graph Laplacian.

simple way to build the graph is to use the k -nearest neighbors approach. Using this approach, for each vertex we add an edge between the vertex and its k -nearest vertices, with k being a method parameter. Distances can be calculated using Euclidean or geodesic distances. A more complex approach is to build a weighted graph using a thresholded Gaussian kernel¹¹ where the weights between vertices i and j are given by

$$W_{i,j} = \begin{cases} \exp\left(-\frac{\text{dist}^2(i,j)}{2\theta^2}\right) & \text{if } \text{dist}(i,j) \leq \kappa \\ 0 & \text{otherwise,} \end{cases} \quad (1)$$

where κ is the threshold, θ is a dispersion factor, and $\text{dist}(i,j)$ is the distance—euclidean or geodesical—between vertices i and j . Another alternative is to build the graph using a data-driven approach based on the estimation of covariance matrices.^{18,19}

3. GSP AND EEG

There is a long tradition of using graph theory in neuroimaging in general²⁰ and EEG in particular.^{21,22} However, it is essential to highlight that this use of graph theory has been circumscribed to analyze the graph obtained from connectivity structures in the brain after applying other processing steps. In this respect, GSP—that considers the analysis of *signals* defined over a graph—is fundamentally different from this classic approach.

The capacity of GSP to deal with signals over graphs has been exploited in several works related to neuroimaging. In general, these works are based on representing the interconnection among brain regions as a graph. In this graph, each brain region is a vertex. The connectivity between vertices is inferred through the anatomical connectivity or the functional relationship of the regions.^{23,24} The signal over the graph is the activity level of each region, determined typically by functional Magnetic Resonance Imaging (fMRI) or similar techniques. For example, in Refs. 25–27, the notion of graph signal frequency is used to characterize adaptability levels and exposure to a task in a subject, using fMRI. Another example is using graph wavelet analysis to find patterns of brain activity and to decompose fMRI data.^{28–30} Using GSP, it is also possible to combine fMRI with diffusion MRI in a multilayer-graph framework that captures transient networks of spatio-temporal connectivity.³¹ As a final example, we mention the use of GSP to classify brain graph signals to understand the etiology of mental diseases.^{32–35}

Presently, most uses of GSP are based mainly on fMRI data^{25,36} and the corresponding graph signal induced by the functional and anatomical connectivity. There are a limited number of works that use GSP with EEG data. In Ref. 37, the authors use GSP to detect epilepsy from the EEG. In this work, the graph is constructed from the time series data using a visibility graph. In Ref. 38, the authors also built the graph from the time series data and showed how using GSP for dimensionality reduction improves the classification performance over classic techniques. Other works used GSP for noise suppression, dimensionality reduction, and classification^{39–41} of EEG signals. All these works ignore geometrical information to build the graph. To the best of our knowledge, the work of Mortaheb et al.⁴² is the only one that uses geometric information by considering the electrodes’ distance to build the graph. GSP is then used to compute the total variation of the graph signal to characterize disorders of consciousness.

4. IMPUTING MISSING EEG DATA USING GSP

In this section we show an experiment where we use GSP to impute the Event-Related Potential (ERP) of a missing electrode based on the ERP of the remaining electrodes. In particular, we do data imputation by solving the graph linear regression problem^{11,43}

$$\begin{aligned} \hat{f} &= \arg \min_x x^T \mathcal{L}x \\ \text{s.t. } f_M &= Mx, \end{aligned}$$

where $\hat{f}$ is the graph signal we wish to estimate, $\mathcal{L}$ is the graph Laplacian, f is the graph signal with missing data, f_M is a vector with the non-missing values, and M is a masking matrix that extracts the non-missing

values. We considered two alternatives for building the graph, one using the k -nearest neighbors approach and another using the thresholded Gaussian kernel.

We used the sample dataset provided by the MNE-Python library.⁴⁴ This dataset consists of a 60-channel electrode recording of the EEG (with one electrode as reference), while the subject was presented with different visual and auditory stimuli.[§] We computed the ERP for the left auditory condition at 100 milliseconds using the average reference (Figure 1-(a)). We considered each electrode as a graph vertex and the value of the ERP at 100 milliseconds on each electrode as the graph signal. In this experiment, we assumed that data from electrode 6 (marked with a red circle in figure 1-(c)) was missing. We used the actual ERP value of this electrode as ground truth to compute the imputation error.

First, we computed the graph using the k -nearest neighbors method. For each vertex, we added an edge between this vertex and its k -nearest neighbors (based on the Euclidean distance of the position of the electrodes). Figures 1-(b) and 1-(c) show the corresponding adjacency matrix of this graph and a plot of the graph and the graph signal, respectively, for a graph built using the $k = 5$ nearest neighbors.

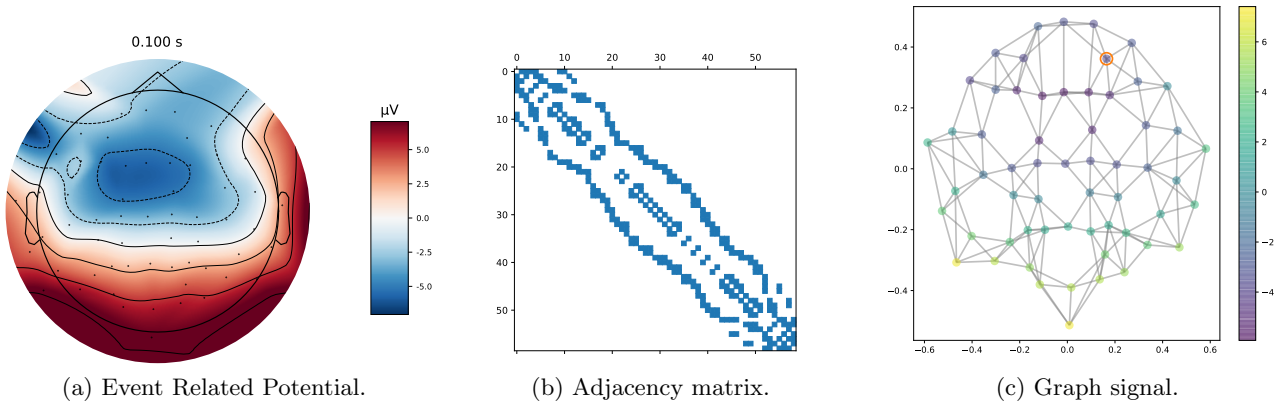


Figure 1: A graph is built using each electrode as a vertex and an edge between each vertex and its five nearest neighbors. The ERP at each electrode (vertex of the graph) is the graph signal: **(a)** Topography of the ERP at 100 milliseconds. **(b)** The adjacency matrix of the graph. **(c)** Graph and signal graph. The red circle marks the electrode with missing data.

To evaluate the performance of the GSP-based imputation method, we computed the magnitude of the error for different values of numbers of neighbors k used to build the graph. We compared this error with the error obtained by imputing the missing data using a standard Radial Basis Function (RBF) interpolator.⁴⁵ Figure 2-(a) shows the result.

To study the effect of using a weighted graph we repeated the previous procedure using a graph built using the thresholded Gaussian kernel (see Eq. (1)). We computed the magnitude of the error for different values of the dispersion parameter θ of the kernel. Figure 2-(b) shows the result. As a reference, we show in the figure the RBF error (red dashed line) and the smallest error obtained with the unweighted graph (purple dashed line).

5. DISCUSSION AND CONCLUSION

The results from the previous section highlight several facts. First, for the results corresponding to the k -nearest neighbors graph, we observe that: (a) For a reasonably built graph, GSP-based data imputation is better than the standard RBF imputation. By reasonably, we mean a graph built with a relatively small number of neighbors. This characteristic makes sense since, at some point, by increasing the number of neighbors k , the GSP imputation begins to consider vertices that are too far away (in a Euclidean distance sense). (b) This effect is amplified by the fact that we are using an unweighted graph, highlighting the relevance of the graph topology on the performance of GSP methods. Secondly, we observe that using a weighted graph built using the

[§]See more details about the dataset in <https://bit.ly/3dzZnHo>.

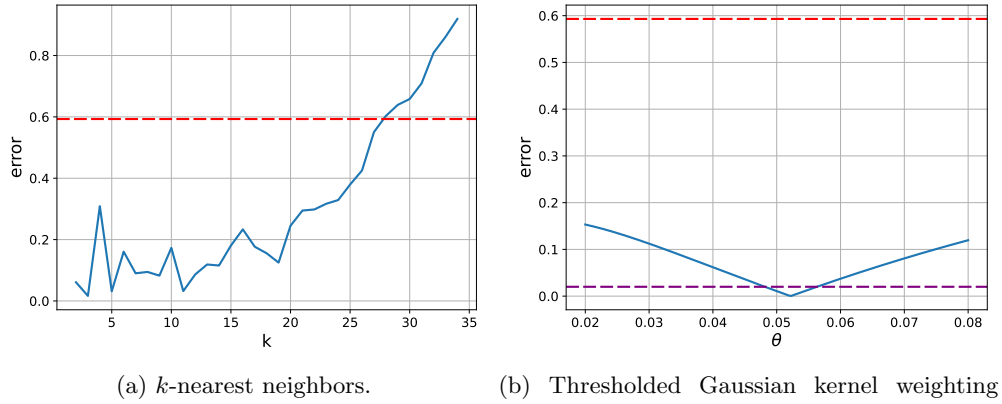


Figure 2: Magnitude of the imputation error. **(a)** With an unweighted graph built using the k -nearest neighbors as a function of k . The red line shows the imputation error when using the classic RBF interpolation method. We observe that when the graph is built using a small number of neighbors, the GSP imputation error is smaller than when using RBF. We also observe that eventually, as k increases, performance deteriorates. This behavior makes sense since the graph is unweighted, and a large value of k implies that the structure of the graph does not capture the spatial relationship among the electrodes. **(b)** With a weighted graph built using a thresholded Gaussian kernel as a function of θ . As before, the red line shows the imputation error when using the RBF interpolation method. The purple line shows the smallest imputation error when using a graph built using the k -nearest neighbors. We observe that the error is uniformly smaller than when using RBF. We also observe that for a range of values of θ , this error is smaller than the one obtained using an unweighted graph. This result is an example where GSP allows us to improve a classic technique. It also shows the importance of building the underlying graph appropriately.

thresholded Gaussian kernel makes it possible to obtain errors smaller than with the unweighted graph. We also observe that the dependency of the error with respect to the kernel scaling parameter is more robust than the dependency of parameter k for the k -nearest neighbor case.

In general, our results confirm that exploiting the intrinsic graph structure over which the EEG signal is defined makes it possible to get better results compared to classic approaches. This, combined with the fact that, as stated in Section 3, the use of GSP to handle EEG signals has received limited attention in the literature, indicates that the use of GSP to process EEG signals is a promising development area. For future works related to the problem of imputing missing EEG data, we suggest performing an exhaustive study of the solution proposed in this work, including analysis over all electrodes, the use of different datasets with a different number of electrodes, the use of data-driven techniques to build the graph, among others.

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